



**DESIGN AND IMPLEMENTATION OF DOCUMENT ARCHIVING
AND SHARING SYSTEM**

**(A CASE STUDY OF NATIONAL PRIMARY HEALTH CARE
DEVELOPMENT AGENCY, ABUJA)**

BY

USMAN OHAGENYI ABUBAKAR

CPC\ABUJA\205807

**SUBMITTED TO COMPUTER PROFESSIONALS REGISTRATION
COUNCIL OF NIGERIA, IN PARTIAL FULFILMENT OF A
MEMBERSHIP OF CPN.**

MARCH, 2026

TITLE PAGE

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DECLARATION

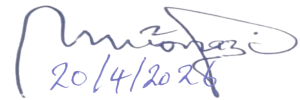
I hereby declare that this Project, titled "**Design and Implementation of Document Archiving and Sharing System (A Case Study of National Primary Healthcare Development Agency, Abuja)**," is the result of my original research work and has not been submitted for any other qualification elsewhere. Any information obtained from other sources, whether published or unpublished, has been duly acknowledged.

CERTIFICATION

This is to certify that the project titled, "**Design and Implementation of Document Archiving and Sharing System (A Case Study of National Primary Healthcare Development Agency, Abuja),**" by **Usman Ohagenyi Abubakar** with Registration Number **CPC\ABUJA\205807** was supervised and approved as one of the requirements for the Membership of the CPN. However, the student bears full responsibility for the work.

Dr. Onazi Omachi Sunday

Supervisor



20/4/2020

Signature & Date

External Examiner

Signature & Date

DEDICATION

This project is dedicated to **Almighty God for His guidance, wisdom, and strength throughout the course of this study.**

It is also dedicated to **my family, mentors, and colleagues** whose encouragement, support, and motivation made the successful completion of this project possible.

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to **God Almighty** for granting me the strength, knowledge, and perseverance required to complete this project successfully.

My profound appreciation goes to my **project supervisor**, whose guidance, constructive criticisms, and valuable suggestions greatly contributed to the successful completion of this research work.

I also wish to acknowledge the support of the **Computer Professionals Registration Council of Nigeria (CPN)** for providing professional standards that guide the development of projects in the field of computing.

Special appreciation goes to the staff and colleagues who provided useful information and encouragement during the development of this project.

Finally, I extend my heartfelt gratitude to **my family and friends** for their continuous support, understanding, and encouragement throughout the course of this work.

ABSTRACT

Effective document management is essential for improving organizational efficiency, ensuring secure information storage, and facilitating collaboration among departments. Many public sector organizations, including the National Primary Health Care Development Agency (NPHCDA), still rely on manual or semi-manual document management processes that are inefficient, time-consuming, and prone to errors such as document loss, duplication, and unauthorized access.

*This project presents the design and implementation of a **Document Archiving and Sharing System** aimed at improving the management of organizational documents within NPHCDA. The proposed system provides a centralized digital platform that enables users to upload, store, retrieve, and share documents across departments while maintaining strict access control and document security.*

*The system was developed using modern web technologies including **React.js** for the user interface, **Node.js with Express.js** for backend services, and **MySQL** as the database management system. The system supports features such as document upload, document categorization, document search, document sharing, access request management, and activity logging.*

*The development of the system followed the **System Development Life Cycle (SDLC)** methodology which includes system analysis, system design, implementation, testing, and evaluation. Data used in the development of the system were collected through observation, interviews, and review of existing document management practices within organizations.*

The implemented system improves document accessibility, reduces document retrieval time, enhances data security, and promotes effective collaboration

among departments. The study concludes that adopting a computerized document archiving and sharing system significantly improves document management processes in public sector organizations.

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CHAPTER ONE

1.0 INTRODUCTION

There has also been a tremendous growth in the amount of Information and Communication Technology available to assist organizations in the development, generation, storage, management, and retrieval of their documentation. The National Primary Health Care Development Agency produces very large amounts of documentation daily, which includes policy documents, reports, guidelines, and operational records. The ineffective management of the documentation using manual methods or disparate systems may cause inefficiencies, data duplication, version conflicts, lack of authorization for file access, loss of records and no accountability for the records.

A Document Achieving and Sharing System (DASS) provides a structured approach to storing, organizing, retrieving, and securing electronic documents. An effective DASS ensures proper version control, access management, and traceability of user actions. This project focuses on the development of a **Secure Departmental Document Achieving and Sharing System with Version Control and Audit Logging**, designed to address the challenges of traditional document handling in organizations.

1.1 BACKGROUND OF THE STUDY

The NPHCDA is responsible for developing and implementing primary healthcare policies and programs across Nigeria. The sensitive and critical nature of health-related documents means there is a pressing need for an efficient system to archive and share these documents securely and reliably, with the use of version controls/audit logs/controlled access etc.

1.2 PROBLEM STATEMENT

Although it is essential to handle documents effectively, NPHCDA is having challenges related to Document Duplication, Version Control, Access to the most recent version of a document and Accountability of Document Action. These issues pose the following challenges:

- i. Inability to track document versions has created confusion about which document is the latest and greatest.
- ii. The potential for document duplication is high, creating issues with space availability and inconsistency between the two versions.
- iii. Absence of an adequate method for tracking the identities of individuals who have accessed, changed or uploaded data.
- iv. Inadequate access control policies that make sensitive documents vulnerable to unauthorized access.
- v. Difficulty tracking and managing the process of auditing document activity to meet compliance or quality assurance or for investigation purposes.
- vi. Risk of document loss or corruption without a centralized and secure storage system.

These issues highlight the need for a centralized, secure, and well-structured document management solution that supports version control, duplicate detection, and audit logging.

1.3 AIM AND OBJECTIVES OF THE STUDY

The main goal of this project is to create and put into use a safe electronic filing system, to store and access records efficiently, which will produce an auditable record of activity for all documents, that can be used by the NPHCDA. Some of the objectives will be:

- i. Create an integrated system to allow for uploading and administrating all the Organization's documents.
- ii. Track and manage document changes with a document versioning control process.
- iii. Prevent duplication through the use of file hashes.
- iv. Monitor activities of users with an audit logging procedure.
- v. Audit logging should include the following information about the user: name, what action they performed, IP address, the device used, and the date/time of the action as well as a record of who uploaded and approved the document.
- vi. Provide a user-friendly interface for uploading, integrating, and retrieving documents.
- vii. Ensure secure storage and integrity of documents within the system.

1.4 SCOPE OF THE STUDY

This project is to create a document archive & sharing solution for the various Departments within the NPHCDA. The document archive & sharing functionality will include:

- i. User authentication & authorization.

- ii. Department based document organization.
- iii. Uploading of new documents & uploading new versions of existing documents.
- iv. Automatic document version number generation.
- iv. Detecting duplicate documents using a cryptographic hashing algorithm.
- v. Audit logging of all critical user actions.
- vi. Secure storage of document metadata and files.
- vii. Viewing and tracking document version history.

Advanced features like document editing in the browser, integration with outside cloud storage, and mobile app support are not offered by the system. The scope of this project is limited to a web-based platform available for use on desktops and laptops only.

1.5 SIGNIFICANCE OF THE STUDY

This study is significant in the following ways:

- i. It provides a practical solution to document achieve challenges faced by NPHCDA.
- ii. It improves accountability and transparency through audit logging.
- iii. It enhances data security by implementing access control and version control mechanisms.
- iv. It demonstrates the application of software engineering principles in solving real-world problems.
- v. It serves as a reference material for future research and development in document management systems.
- vi. It fulfils the academic and professional requirements of the Computer Professionals (Registration Council) of Nigeria (CPN).

1.6 SYSTEM ARCHITECTURE

The system follows a **three-tier architecture**, consisting of the presentation layer, application layer, and data layer.

1.6.1 PRESENTATION LAYER (FRONTEND)

- i. Developed using **React.js**
- ii. Provides user interfaces for login, document upload, version selection, and document preview
- iii. Handles form validation and user interaction
- iv. Communicates with the backend via RESTful APIs

1.6.2 APPLICATION LAYER (BACKEND)

- i. Developed using **Node.js and Express.js**
- ii. Handles business logic such as authentication, document upload, version control, and duplicate checking
- iii. Generates file hashes for duplicate detection
- iv. Records audit logs for all system actions
- v. Manages file renaming using timestamps and version numbers

1.6.3 DATA LAYER (DATABASE & STORAGE)

- i. Uses **MySQL** for structured data storage
- ii. Stores document metadata, version history, user details, and audit logs
- iii. Stores physical document files on the server filesystem
- iv. Ensures data integrity and transactional consistency

CHAPTER TWO

REVIEW OF RELATED LITERATURE

2.1 INTRODUCTION

This chapter reviews existing literature related to document archiving systems, version control mechanisms, audit logging, and secure information systems. The review examines concepts, technologies, and related works to establish a foundation for the proposed system and to justify its design choices

2.2 CONCEPT OF DOCUMENT ARCHIVING SYSTEM

A Document Archiving System (DAS) is a computerized system used to store, manage, track, and control electronic documents. According to Laudon and Laudon (2019), a DMS enables organizations to efficiently organize digital information while ensuring accessibility, security, and integrity.

Modern DAS solutions provide features such as document categorization, metadata tagging, version control, access permissions, and audit trails. These features reduce redundancy, improve collaboration, and enhance operational efficiency.

2.3 DOCUMENT VERSION CONTROL

Document control refers to the systematic management of changes made to documents over time. It allows multiple versions of a document to exist while maintaining a record of modifications and ensuring that users can identify the most current version.

Pressman (2020) describes version control as a critical component of software and document lifecycle management. Without version control, organizations risk data inconsistency, accidental overwrites, and loss of historical records.

Common versioning techniques include:

- i. Sequential version numbering (e.g., v1.0, v1.1)
- ii. Change notes for tracking modifications
- iii. Marking the active version

The proposed system implements automatic version incrementing and retains historical versions for traceability.

2.4 DUPLICATE DOCUMENT DETECTION

Duplicate documents create confusion, waste storage resources, and compromise data integrity. One effective method of duplicate detection is file hashing.

Cryptographic hash functions such as SHA-256 generate a unique fixed-length hash for each file. According to Schneier (2018), even a minor change in a file produces a completely different hash value, making hashing reliable for duplicate detection.

In this project, hashing is used to:

- i. Detect exact duplicate uploads
- ii. Prevent redundant document versions
- iii. Maintain system integrity

2.5 AUDIT LOGGING IN INFORMATION SYSTEMS

Audit logging is the process of recording system activities to ensure accountability, transparency, and security. Audit logs capture details such as user identity, action performed, time, IP address, and device information.

Stallings (2021) emphasizes that audit trails are essential for:

- i. Security monitoring
- ii. Regulatory compliance
- iii. Incident investigation
- iv. User accountability

The proposed system maintains a comprehensive audit log that records every critical action performed by users.

2.6 SECURITY AND ACCESS CONTROL

Security is a critical requirement for document archiving systems. Access control mechanisms ensure that only authorized users can view or modify documents.

Role-Based Access Control (RBAC) is widely used in enterprise systems. It assigns permissions based on user roles and departments. According to Sandhu et al. (2019), RBAC improves security while simplifying access management.

This project applies authentication and department-based authorization to restrict access to documents.

2.7 REVIEW OF RELATED SYSTEMS

Several existing systems provide document management solutions:

1. Google Drive

- Supports document storage and versioning
- Limited audit transparency for organizations

2. Microsoft SharePoint

- Enterprise-level document management
- High cost and complex configuration

3. Dropbox Business

- Cloud-based file management
- Limited customization for departmental workflows

Despite their strengths, these systems may not fully meet the specific needs of departmental document management with customized audit logging, as required in many organizations.

2.8 SUMMARY OF LITERATURE REVIEW

The literature reviewed highlights the importance of structured document management, version control, duplicate detection, and audit logging in modern information systems. Existing solutions either lack customization or are costly to deploy. This project addresses these gaps by providing a secure, department-focused document archiving system with comprehensive version control and audit logging.

CHAPTER THREE

SYSTEM ANALYSIS AND DESIGN

3.1 INTRODUCTION

System analysis and design involve studying the existing operational procedures of an organization and developing improved information systems that solve identified problems. It focuses on understanding the requirements of users, identifying system limitations, and designing solutions that improve efficiency, security, and performance.

In many public health institutions such as the National Primary Health Care Development Agency (NPHCDA), document management processes are still largely dependent on manual filing systems and fragmented digital storage. These traditional methods often result in document loss, duplication of information, delays in document retrieval, and limited collaboration among departments.

According to recent studies, the transition from paper-based documentation to digital document management systems significantly improves information accessibility, operational efficiency, and regulatory compliance within

organizations. Digital document systems provide secure storage, quick retrieval, version control, and centralized access to documents across departments.

This chapter examines the existing document management system used within NPHCDA, identifies its limitations, and proposes a new computerized document archiving and sharing system that improves document storage, retrieval, and collaboration among departments.

The flow Chart of the existing system

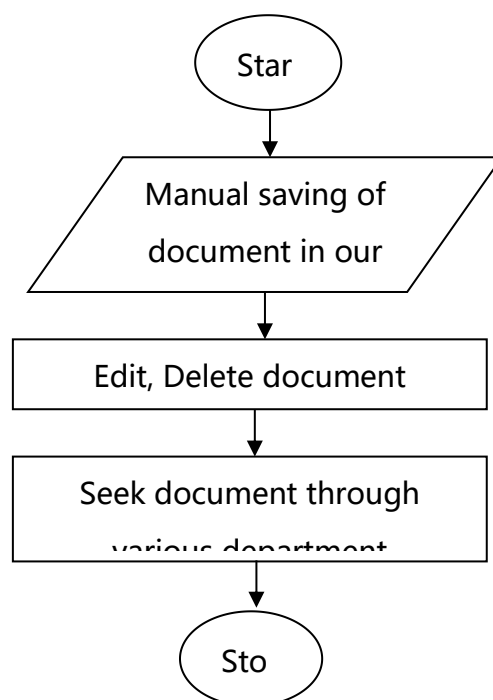


Figure 1.0: Flow Chart of the existing former System

3.2 ANALYSIS OF THE EXISTING SYSTEM

The existing document management system within many public health organizations such as NPHCDA primarily relies on a **manual or semi-manual process** for document storage and retrieval.

The process typically involves the following steps:

1. Documents are created manually or through office applications.
2. Printed copies are stored in physical files or cabinets.
3. Documents are labelled and categorized manually.
4. Staff retrieve documents through manual searching.
5. Document sharing between departments occurs through physical transfer or email attachments.

While some documents may be stored digitally, these files are often saved in scattered locations such as personal computers, USB drives, or local office servers. This decentralized approach leads to data fragmentation and inefficient document tracking.

Research shows that organizations using manual document management processes often experience delays in document retrieval, data duplication, and difficulties maintaining accurate records.

Furthermore, manual document storage makes it difficult to maintain proper document retention policies and regulatory compliance.

3.3 MERITS OF THE EXISTING SYSTEM

Despite its limitations, the current manual document management system possesses certain advantages.

1. Simplicity of Operation

The existing system does not require complex technological infrastructure. Staff can easily understand and operate manual filing systems without extensive training.

2. Low Initial Implementation Cost

Manual systems require minimal technological investment since they rely primarily on physical storage such as filing cabinets and paper documents.

3. Familiarity with Staff

Many administrative personnel are accustomed to traditional record management systems, making the process easier to adopt.

4. Limited Dependence on Technology

Manual systems can still function even during network failures or computer system breakdowns.

3.4 LIMITATIONS OF THE EXISTING SYSTEM

The current document management system within NPHCDA presents several challenges.

1. Difficulty in Document Retrieval

Searching for documents manually is time-consuming and inefficient, especially when dealing with large volumes of records.

2. Risk of Document Loss

Paper-based systems are vulnerable to damage, theft, fire, and misplacement.

3. Limited Document Sharing

Manual document sharing between departments delays decision-making processes and reduces collaboration.

4. Poor Security Control

Physical documents are difficult to secure and control. Unauthorized individuals may access sensitive records.

5. Lack of Document Tracking

Manual systems do not provide audit trails to monitor document access or modifications.

6. Storage Constraints

Large quantities of paper records require extensive physical storage space.

Research indicates that organizations implementing digital document management systems experience significant reductions in document retrieval time and storage costs.

3.5 JUSTIFICATION OF THE NEW SYSTEM

The proposed Document Archiving and Sharing System for NPHCDA is designed to address the limitations of the existing system.

The new system provides several improvements:

1. Centralized Document Repository

All documents will be stored in a centralized database that allows authorized users to access them easily.

2. Faster Document Retrieval

Digital indexing and search functionality allow users to locate documents quickly.

3. Improved Security

The system will include authentication, authorization, and audit logging to ensure that only authorized users access sensitive documents.

4. Efficient Document Sharing

Departments will be able to share documents electronically through the system.

5. Document Version Control

The system will maintain multiple versions of documents to prevent loss of information.

6. Reduced Paper Usage

Digital archiving reduces dependency on physical paper storage.

Studies show that digital document management systems significantly improve operational efficiency, compliance, and collaboration within healthcare institutions.

3.6 DEMERITS OF THE NEW SYSTEM

Although the proposed system offers several advantages, it also has some limitations.

1. Initial Implementation Cost

Developing and deploying the system requires financial investment in hardware, software, and infrastructure.

2. Training Requirements

Staff members must be trained to effectively use the new system.

3. Dependence on Technology

The system requires reliable electricity and internet connectivity.

4. Cybersecurity Risks

Digital systems may be vulnerable to cyber threats if adequate security measures are not implemented.

However, these limitations can be mitigated through proper system maintenance, cybersecurity protocols, and regular staff training.

3.7 METHODOLOGY

The development of the proposed system follows the **System Development Life Cycle (SDLC)** methodology.

The SDLC provides a structured approach for system development and includes the following phases:

1. System Planning
2. System Analysis
3. System Design
4. System Development
5. System Testing
6. System Implementation
7. System Maintenance

The methodology ensures that the system meets user requirements and functions effectively within the operational environment.

6 Phases of the Software Development Life Cycle



3.8 DATA COLLECTION

Data for the development of the proposed system was collected using the following methods:

1. Oral Interview

Interviews were conducted with NPHCDA administrative staff and record management personnel to understand the current document management procedures.

2. Observation

Direct observation of document handling procedures within the organization helped identify inefficiencies in the existing system.

3. Document Review

Existing documents, reports, and record management policies were analyzed to understand document storage requirements.

4. Literature Review

Academic journals and previous research on digital document management systems were reviewed to identify best practices and technological approaches.

These data collection methods provided valuable insights into the challenges faced by the organization and helped define system requirements.

3.9 THE PROPOSED NEW SYSTEM

The proposed system is a **web-based Document Archiving and Sharing System** designed to manage digital records within NPHCDA.

The system will allow authorized users to:

- Upload documents

- Archive documents
- Search documents
- Share documents between departments
- Request document access
- Track document history

The system will also support role-based access control where different users such as administrators, department officers, and staff members have different permissions.

Research shows that electronic document management systems improve collaboration and workflow efficiency in healthcare institutions.

3.10 REQUIREMENT SPECIFICATION

The requirements for the proposed system are divided into two (2). These are Hardware requirements and software requirements.

3.10.1 HARDWARE REQUIREMENTS

The hardware required for the system includes:

- Computer systems
- Server system
- Network router
- External storage devices
- Backup power supply (UPS)

Minimum hardware specifications include:

- Processor: Intel Core i5 or equivalent
- RAM: 8GB

- Storage: 500GB Hard Drive
- Network connectivity

3.10.2 SOFTWARE REQUIREMENTS

The software components required for the system include:

- Operating System (Windows or Linux)
- Web Browser (Chrome, Edge, Firefox)
- Database Management System (MySQL)
- Web Server (Node.js / Express)
- Frontend Framework (React)
- Programming Language (JavaScript)

The system will operate through a web-based interface accessible via internet or local network.

3.11 ARCHITECTURAL DESIGN OF THE PROPOSED SYSTEM

The proposed system follows a **three-tier architecture**.

1. Presentation Layer

The presentation layer provides the user interface for interacting with the system. It includes web pages that allow users to upload, search, and share documents.

2. Application Layer

This layer contains the system logic that processes user requests and manages system operations such as document storage, authentication, and access control.

3. Data Layer

The data layer manages the database that stores documents, user information, and system logs.

The architecture ensures scalability, maintainability, and security.

Modern digital document systems often use distributed architectures to improve data security and information sharing between healthcare organizations.

CHAPTER FOUR

SYSTEM IMPLEMENTATION

4.1 INTRODUCTION

System implementation involves the practical realization of the proposed system design. It includes the development of software modules, database structures, user interfaces, and system integration required to achieve the objectives of the document archiving and sharing system.

The implementation of the **Document Archiving and Sharing System for NPHCDA** was carried out using modern web technologies to ensure efficiency, scalability, and security. The system enables users to upload, archive, search, share, and request access to documents within the organization.

The system was developed using the following technologies:

- **Frontend:** React.js
- **Backend:** Node.js with Express.js

- **Database:** MySQL
- **Authentication:** JSON Web Tokens (JWT)
- **API Communication:** Axios

These technologies enable the development of a responsive, secure, and scalable document management platform.

4.2 SYSTEM DESIGN

The design of the proposed system includes several tools such as Context Diagrams, Level 0 Data Flow Diagrams (DFD), Use Case Diagrams, and Entity Relationship Diagrams (ERDs). These tools help in defining the functional and structural aspects of the system. The ER diagram is particularly useful in illustrating the relationships between entities and their functionalities.

The system development follows the Prototype Methodology, which ensures that user requirements are effectively incorporated into the system design. This iterative approach allows for continuous feedback and refinement.

4.3 SYSTEM TESTING

4.3.1 Component Testing

Component testing was conducted in the development environment (Microsoft Visual Studio Code) immediately after the code was written. Each individual function was compiled and executed to ensure it performed the intended tasks. If any section of the code failed to meet the required outcome, it was reviewed against the specification and corrected accordingly.

4.3.2 Bottom-Up Integration

Test Integration testing involved combining all components of the system to ensure they function correctly as a whole. During this phase, errors were identified and rectified. Unit testing was also revisited to pinpoint issues and ensure system stability. According to Sommerville (2001), integration testing should ideally be performed on a separate machine from the one used during development.

4.4 System Requirements Analysis

A comprehensive system analysis was conducted to define the requirements for system development. These requirements are categorized into user requirements, functional requirements, and non-functional requirements.

4.4.1 User Requirements

The system should allow users to:

- i. Upload and update information about documents in each department.
- ii. View and search for details of users, documents, and staff request.
- iii. Subscribe or log in to the system anytime.

4.4.2 Functional Requirements

- i. The system should perform the following functions:
- ii. Allow users to change passwords for security purposes.
- iii. Require users to create a password of at least five characters when changing passwords.
- iv. Enable users to record all information about document upload.
- v. Allow bulk user registration using the import feature

4.4.3 Non-Functional Requirements

The system should meet the following non-functional requirements:

- i. Be easy to install and configure.
- ii. Occupy minimal disk space for efficient storage management.
- iii. Have a user-friendly interface that is easy to learn and use.
- iv. Be platform-independent for wider usability.

4.4.4 Software Requirements

The system requires the following software:

- i. Windows 7, Windows 8, or Windows 10 operating system.
- ii. Relational Database Management System (RDBMS) for efficient data handling.

4.4.5 Hardware Requirements

The system's minimum hardware requirements include:

- i. A CPU with at least a Pentium 4 processor.
- ii. A minimum of 512MB RAM.
- iii. A hard disk with at least 500GB storage capacity.

4.5 SYSTEM FLOWCHART

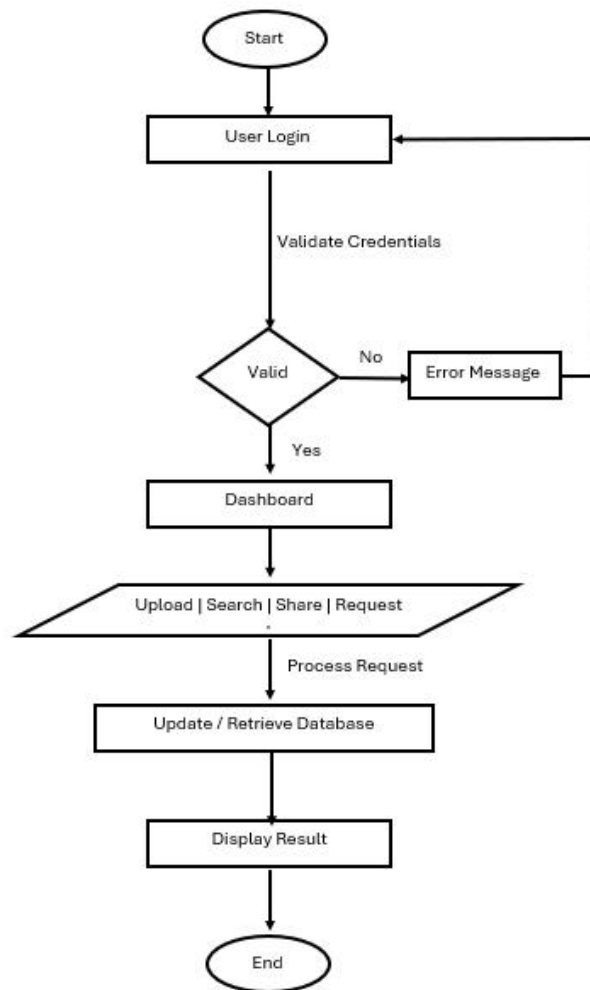
A system flowchart represents the sequence of operations carried out by the system from the moment a user interacts with the application until the requested operation is completed.

System Flowchart Description

The document archiving system follows this operational flow:

1. User accesses the system.
2. User logs into the system.
3. System validates user credentials.
4. User selects system operation:
 - Upload document
 - View documents
 - Search documents
 - Request access
 - Manage documents
5. System processes the request.
6. Database stores or retrieves data.
7. System displays results to the user.

Flowchart Representation



4.6 USE CASE DIAGRAM

The use case diagram illustrates how users interact with the system and what functionalities they can perform.

System Actors

1. Administrator
2. Department User
3. Super Administrator

Administrator Use Cases

- Manage users
- Approved document deletion
- Approve document access requests
- View audit logs

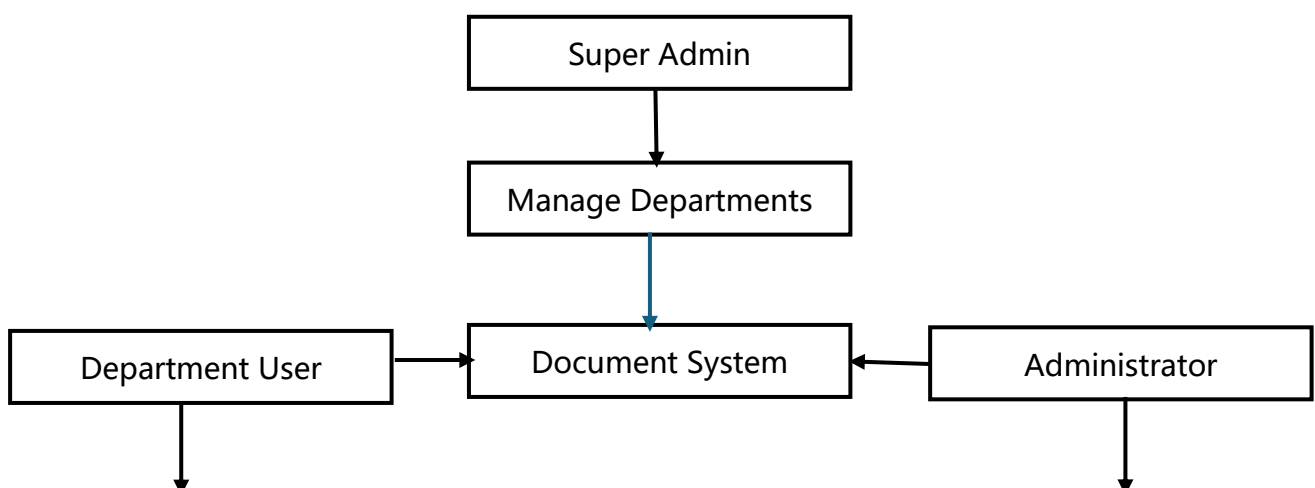
Department User Use Cases

- Upload documents
- Search documents
- View document details
- Request document access
- View shared documents

Super Administrator Use Cases

- Manage departments
- Manage system configuration
- View system logs

Use Case Representation



Upload Document
Search Document
Request Access

Manage Users
Approve Access
View Logs

4.7 DATABASE SCHEMA

The database schema defines how data is stored, organized, and related within the system.

The system uses a **relational database (MySQL)** consisting of multiple tables.

4.7.1 USERS TABLE

Stores user account information.

```
CREATE TABLE users (  
  id INT AUTO_INCREMENT PRIMARY KEY,  
  name VARCHAR (100),  
  email VARCHAR (100),  
  password VARCHAR (255),  
  department_id INT,  
  role ENUM ('admin', 'user', 'superadmin'),  
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP  
)
```

4.7.2 DEPARTMENTS TABLE

Stores departmental information.

```
CREATE TABLE departments (  
  id INT AUTO_INCREMENT PRIMARY KEY,  
  name VARCHAR (100),
```

description TEXT
)

4.7.3 DOCUMENTS TABLE

Stores document metadata.

```
CREATE TABLE documents (  
  id INT AUTO_INCREMENT PRIMARY KEY,  
  title VARCHAR(255),  
  description TEXT,  
  category_id INT,  
  subcategory_id INT,  
  department_id INT,  
  classification ENUM('Public', 'Internal', 'Confidential', 'Restricted'),  
  document_status VARCHAR(50),  
  document_file VARCHAR(255),  
  created_by INT,  
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP  
);
```

4.7.4 DOCUMENT CATEGORIES

```
CREATE TABLE document_categories (  
  id INT AUTO_INCREMENT PRIMARY KEY,  
  name VARCHAR(100)  
);
```

4.7.5 DOCUMENT SUBCATEGORIES

```
CREATE TABLE document_subcategories (  
  id INT AUTO_INCREMENT PRIMARY KEY,  
  category_id INT,  
  name VARCHAR(100)  
);
```

4.7.6 ACCESS REQUESTS TABLE

Stores requests for document access.

```
CREATE TABLE document_access_requests (  
  id INT AUTO_INCREMENT PRIMARY KEY,  
  document_id INT,  
  requested_by INT,  
  reason TEXT,  
  status ENUM('Pending','Approved','Rejected'),  
  reviewed_by INT,  
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP  
);
```

4.7.7 AUDIT LOGS TABLE

Tracks system activities.

```
CREATE TABLE audit_logs (  
  id INT AUTO_INCREMENT PRIMARY KEY,  
  user_id INT,  
  action VARCHAR(100),  
  entity_type VARCHAR(50),  
  entity_id INT,
```

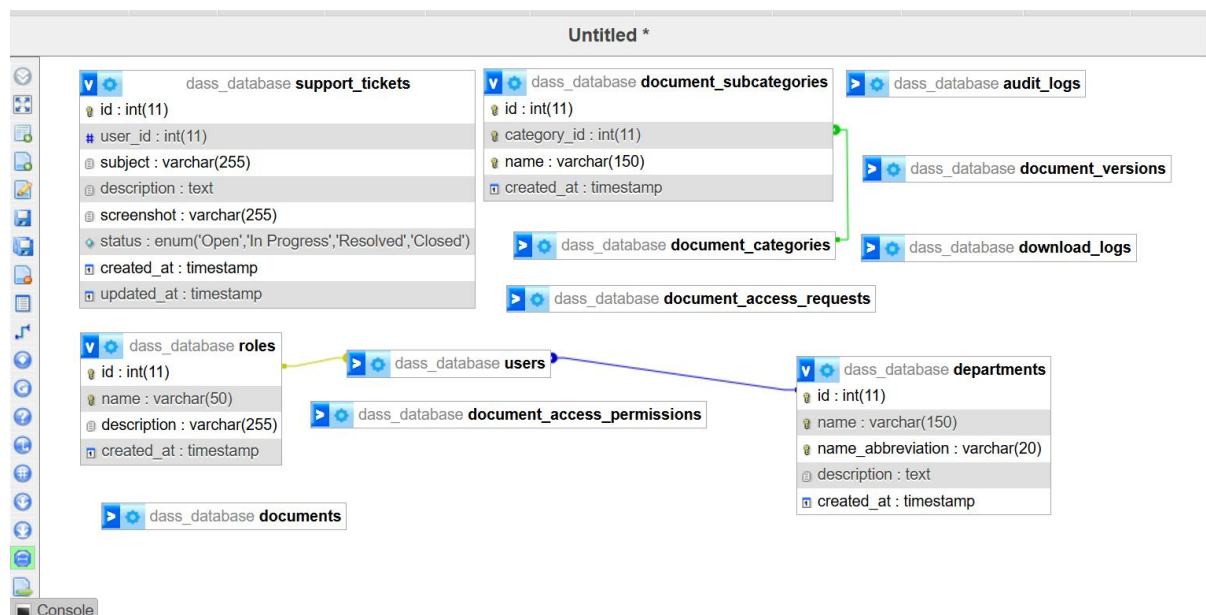
```

description TEXT,
ip_address VARCHAR(50),
created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

```

4.7.8 ENTITY RELATIONSHIP DIAGRAM

The ERD below illustrates all the entities with their relations and associated attributes to makeup the Document Archiving and Sharing System.



4.8 SYSTEM IMPLEMENTATION

4.8.1 Program screenshot

The Login Form

This is what allows users to input their valid login details to allow them to access the system. Login form loads to the real user's interface depending on who has logged in.

Federal Republic of Nigeria

National Primary Health Care Development Agency
Document Archiving & Sharing System (DASS)

Secure platform for managing, archiving and sharing official government documents across departments.

Login Portal

Username
Enter your username

Password
Enter your password

Login to System

© 2026 National Primary Health Care Development Agency

Figure 4.8.1 Login Form

4.8.2 Department Focal Person Dashboard

NPHCDA-DASS
1791

Good Morning
Welcome back, Usman

Dashboard
Planning Research and statistic
Last Login: 3/16/2026, 9:16:49 AM

Total Documents
1,250

Pending Requests
32

Active Users
120

Expiring Documents
7

Recent Activities
Last 7 days

Document	Uploaded By	Department	Status	Date
Annual Report 2025	Jane Doe	PRSDFF	Approved	2026-02-16
Policy Document	John Smith	Finance	Pending	2026-02-14
Partner MoU	Mary Johnson	Partners	Expired	2026-01-30

© 2026 NPHCDA Archive & Sharing System

Figure 4.8.2 Department focal person dashboard

4.8.3 Upload Document

The Department Focal Person (DFP) uses this part of the interface for uploading document into the system.

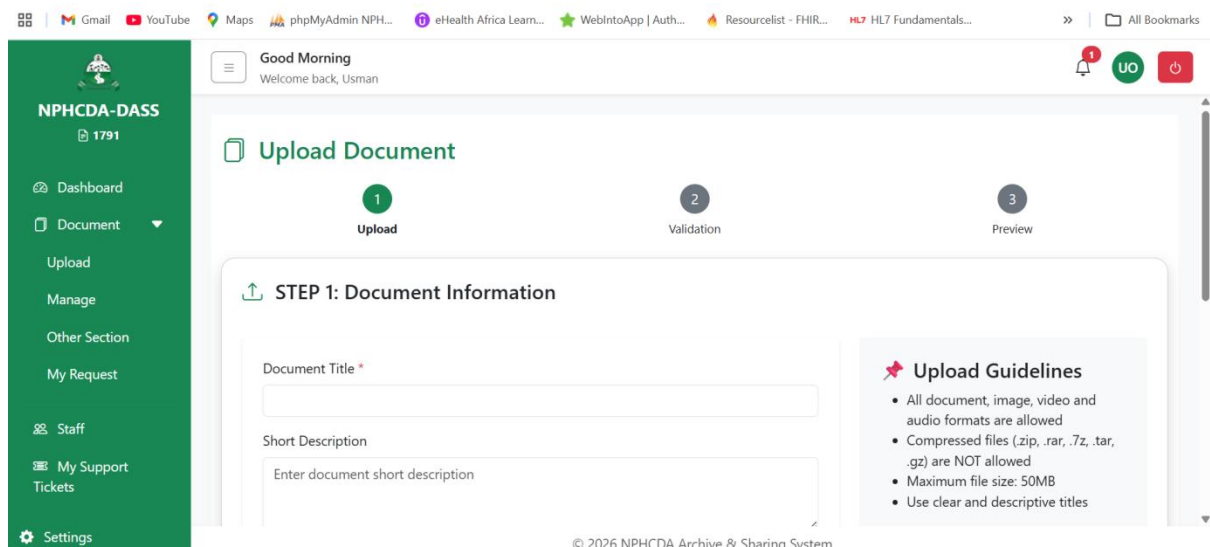


Figure 4.8.3 Upload Document

4.8.4 List Documents

The Department Focal Person (DFP) uses this part of the interface to see all the documents uploaded from his department into the system.

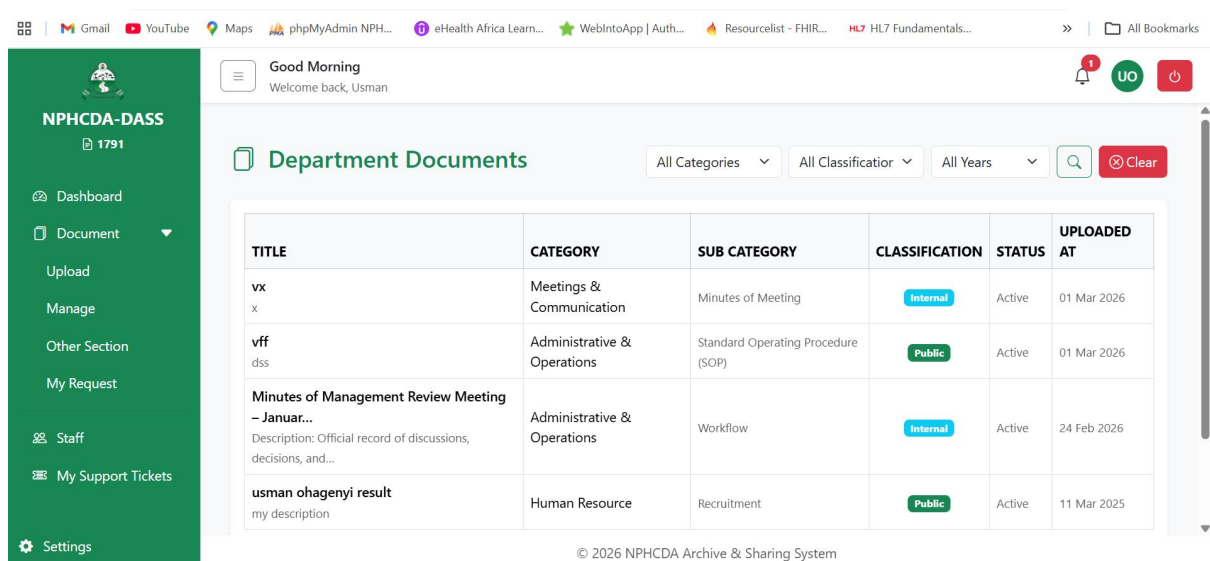


Figure 4.8.4 List Document

4.8.5 Document Detail

The Department Focal Person (DFP) uses this part of the interface for viewing document details.

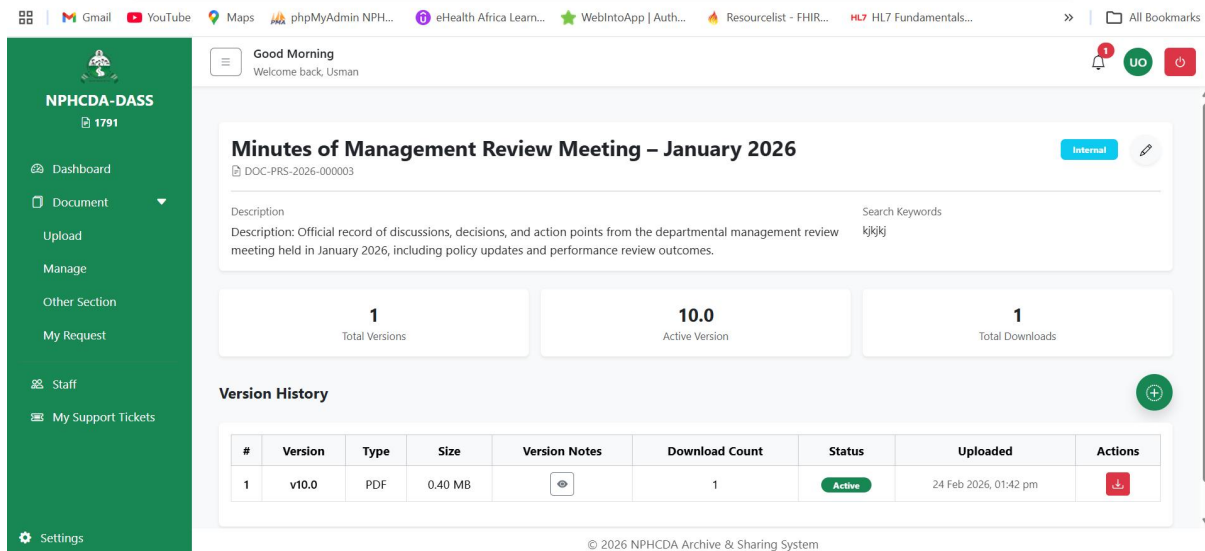


Figure 4.8.5 Document Detail

4.8.6 Document Requested From Other Departments

The Department Focal Person (DFP) uses this part of the interface for viewing all documents requested from other departments to confirm if access is granted for download.

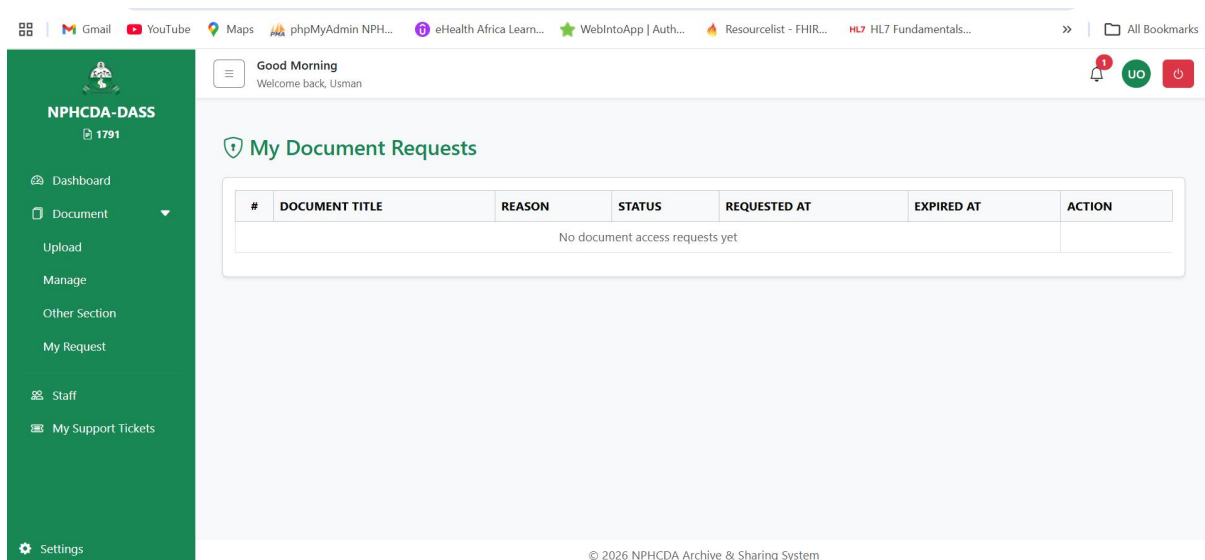


Figure 4.8.6 Document request from other departments page

4.8.7 Registering Staff

The DFP uses this part of the interface for filling in the information of staff under his department into the system.

Add New Staff

Personal Information

Title: Select
Full Name: Enter full name

Gender: Select Gender

Job Information

Designation: e.g Program Officer
File Number: e.g 1739

Contact Information

Email Address: staff@nphcda.gov.ng
Phone Number: 080xxxxxxx

Cancel Save Staff

Figure 4.8.7 Register staff

4.8.8 Staff List

The DFP uses this part of the interface for managing staff in the system.

Department Staff

Showing 2 staff

#	Name	Designation	Email	Phone	Status	Actions
1	Fatima Lukman	Accountant	fati@gmail.com	08042537629	Active	Action
2	Aisha Adamu	Program Analyst 1	aisha@gmail.com	07056369105	Active	Action

Previous Page 1 of 1 Next

Figure 4.8.8 Manage staff

4.8.9 Help Desk

The DFP uses this part of the interface for contacting help desk to report any issues related to the system.

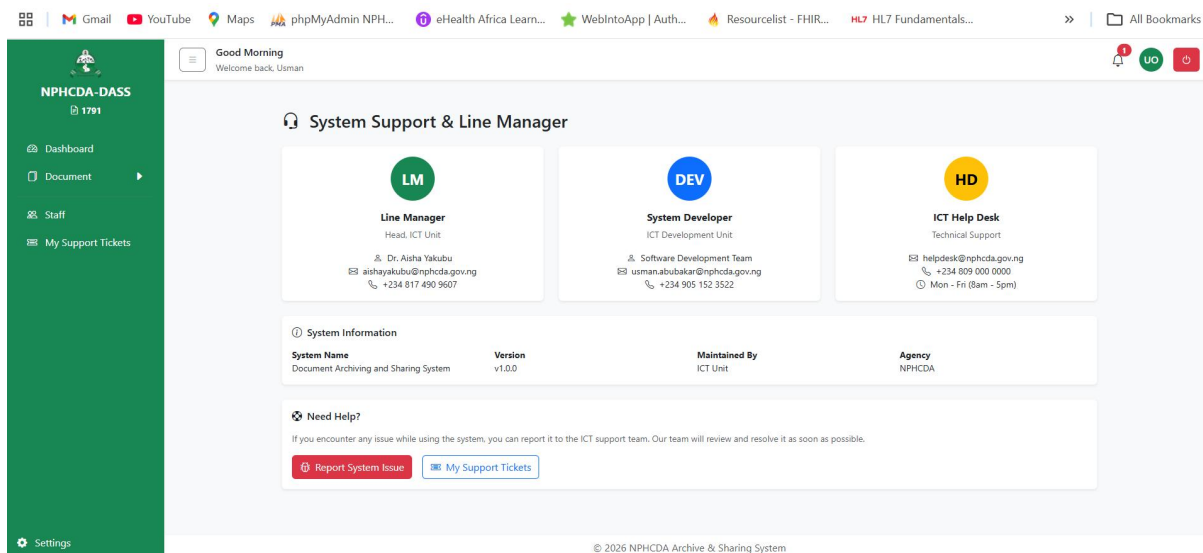


Figure 4.8.9 Manage staff

CHAPTER FIVE

SUMMARY, CONCLUSION AND RECOMMENDATIONS

5.1 SUMMARY

This project focused on the “**Design and Implementation of Document Archiving and Sharing System (A Case Study of National Primary Healthcare Development Agency, Abuja)**,” The objective of the study was to develop a computerized system that enhances the storage, retrieval, management, and sharing of organizational documents within the agency.

The study was motivated by the numerous challenges associated with the existing manual and semi-manual document management systems used in many public sector organizations. The traditional system involved storing documents in physical files, cabinets, and scattered digital locations, which made document retrieval slow and inefficient. Additionally, the absence of centralized storage and document tracking mechanisms often resulted in document duplication, loss of records, unauthorized access, and poor interdepartmental collaboration.

Chapter One of the project introduced the background of the study, highlighting the importance of effective document management in large organizations such as NPHCDA. The chapter also defined the research problem, objectives of the study, scope of the system, significance of the study, and limitations encountered during the research.

Chapter Two presented the **literature review**, which examined previous studies related to document management systems, digital archiving, and information sharing technologies. The review highlighted how modern document management systems help organizations improve operational efficiency, reduce paper-based processes, and enhance data security. The chapter also examines various technological approaches used in the development of document archiving systems, including database management systems, web-based technologies, and digital storage solutions.

Chapter Three discussed system **analysis and design**, which examined the existing document management system within the organization. The analysis identified the strengths and weaknesses of the current system and provided justification for the development of the proposed system. The chapter also described the methodology adopted for the study, methods of data collection, system requirements, and architectural design of the proposed system.

Chapter Four focused on the **system implementation**, which involved the development of the document archiving and sharing system using modern web technologies. The system was implemented using **React.js for the frontend interface, Node.js with Express.js for backend services, and MySQL for database management**. The chapter also presented the database schema, system architecture, user interfaces, and testing procedures used to ensure that the system functions effectively.

The implemented system provides several important functionalities including:

- Uploading and archiving documents
- Searching and retrieving documents
- Managing document classifications
- Sharing documents between departments
- Requesting access to restricted documents
- Tracking document activities through audit logs

Through these features, the proposed system significantly improves the management of documents within NPHCDA and provides a centralized platform for secure document storage and collaboration.

5.2 CONCLUSION

Effective document management plays a critical role in the efficient operation of modern organizations. Public sector institutions such as NPHCDA generate large volumes of administrative and operational documents that must be properly stored, managed, and shared among departments.

The traditional manual approach to document management presents several challenges including limited accessibility, inefficient retrieval processes, document loss, and inadequate security controls. These challenges reduce organizational productivity and increase the risk of information mismanagement.

The Document Archiving and Sharing System developed in this project provides a digital solution that addresses these challenges. The system enables centralized document storage, faster document retrieval, improved document security, and efficient sharing of information across departments.

The implementation of the system demonstrates how modern information technologies can be applied to improve organizational processes and support effective information management. By digitizing document storage and providing controlled access to records, the system enhances transparency, accountability, and operational efficiency within the organization.

Furthermore, the system provides important security features such as user authentication, role-based access control, document classification, and audit logging. These features ensure that sensitive documents are protected from unauthorized access while still allowing legitimate users to retrieve necessary information when required.

The successful implementation of the system confirms that digital document management solutions are essential for organizations seeking to improve efficiency, reduce operational costs, and ensure proper record keeping.

5.3 RECOMMENDATIONS

Based on the findings and results of this study, the following recommendations are proposed:

1. Adoption of the Proposed System

The National Primary Health Care Development Agency should adopt and deploy the proposed Document Archiving and Sharing System to replace or complement the existing manual document management processes. The system will improve efficiency in document storage, retrieval, and sharing.

2. Staff Training

Employees and administrative staff should receive adequate training on how to use the system effectively. Proper training will ensure smooth adoption of the system and reduce resistance to technological change.

3. Implementation of Data Backup Mechanisms

Regular data backups should be implemented to prevent loss of important documents in case of system failure, cyberattacks, or hardware malfunction.

4. Strengthening System Security

To ensure the protection of sensitive organizational documents, the system should incorporate advanced security measures such as:

- Secure authentication methods
- Data encryption
- Access control policies
- Regular security audits

5. Integration with Other Organizational Systems

The document archiving system can be integrated with other information systems within the organization such as health management systems, administrative systems, and reporting platforms to improve data sharing and workflow efficiency.

6. Continuous System Maintenance and Updates

The system should be regularly maintained and updated to fix bugs, improve performance, and introduce new features that meet evolving organizational needs.

7. Implementation of Cloud-Based Storage

Future improvements may include the integration of cloud storage technologies to enhance document accessibility, scalability, and disaster recovery capabilities.

5.4 SUGGESTIONS FOR FURTHER RESEARCH

Future researchers can expand this study by exploring additional features that enhance document management systems, such as:

- Artificial Intelligence for automated document classification
- Blockchain technology for secure document verification
- Machine learning algorithms for intelligent document search
- Cloud-based document collaboration systems

These technologies can further improve the effectiveness of digital document management systems in large organizations.

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PROGRAM SOURCE CODE

```
import React, { useEffect, useState } from "react";  
import axios from "axios";
```

```

import { useNavigate } from "react-router-dom";

const Login = () => {
  // Axios will send cookies (httpOnly JWT)
  axios.defaults.withCredentials = true;

  const [values, setValues] = useState({ username: "", password: "" });
  const [errorMessage, setErrorMessage] = useState(""); // Backend errors
  const [successMessage, setSuccessMessage] = useState(""); // Success

  const [showPassword, setShowPassword] = useState(false);
  const [loggingIn, setLoggingIn] = useState(false);
  const navigate = useNavigate();

  useEffect(() => {
    axios
      .get("http://localhost:3000/verify")
      .then((res) => {
        if (res.data.Status) {
          const { role } = res.data;

          if (role === "ADMIN") navigate(`/admin`);
          else if (role === "FOCAL_PERSON") navigate(`/department`);
          else if (role === "SUPER_ADMIN") navigate(`/superadmin`);
          else if (role === "STAFF") navigate(`/staff`);
        }
      })
  })
}

```

```

        .catch(() => {});
    }, [navigate]);

// Login handler
const handleLogin = async (e) => {
    e.preventDefault();
    setLoggingIn(true);
    setErrorMessage("");
    setSuccessMessage("");

    try {
        const response = await axios.post(
            "http://localhost:3000/auth/login",
            values,
        );

        if (response.data.Status) {
            setSuccessMessage(response.data.Message);

            // Optional: store role/id locally
            //localStorage.setItem("role", response.data.role);
            //localStorage.setItem("id", response.data.id);

            const role = response.data.role;

```

```

// Delay navigation so user sees success message
setTimeout(() => {
  if (role === "SUPER_ADMIN") navigate("/superadmin");
  else if (role === "ADMIN") navigate("/admin");
  else if (role === "FOCAL_PERSON") navigate("/department");
  else if (role === "STAFF") navigate("/staff");
}, 800);
}
} catch (error) {
  // Extract backend error messages properly
  const backendError =
    error.response?.data?.Error ||
    error.response?.data?.message ||
    "An unexpected error occurred. Please try again.";

  setErrorMessage(backendError);
} finally {
  setLoggingIn(false);
}
};

return (
  <div className="vh-100 d-flex flex-column bg-light">
    {/* Top Government Header */}
    <div className="bg-success text-white py-2 text-center shadow-sm">

```


Federal Republic of Nigeria

</div>

<div className="container flex-grow-1 d-flex align-items-center justify-content-center">

<div className="row w-100 justify-content-center">

{/* LEFT SIDE INFORMATION */}

<div className="col-md-6 d-none d-md-flex flex-column justify-content-center text-center">

<h3 className="fw-bold text-success">

National Primary Health Care Development Agency

</h3>

<p className="text-muted">

Document Archiving & Sharing System (DASS)

</p>

<div className="mt-3 small text-muted">

Secure platform for managing, archiving and sharing official government documents across departments.

</div>

</div>

{/* LOGIN CARD */}

<div className="col-md-4">

<div className="card shadow-lg border-0 rounded-4">

<div className="card-body p-4">

{/* Logo */}

<div className="text-center mb-3">

<h5 className="fw-bold text-success mt-2">Login Portal</h5>

</div>

{/* Error Alert */}

{errorMessage && (

<div className="alert alert-danger py-2 text-center">

{errorMessage}

</div>

```
)}
```

```
{/* Success Alert */}
```

```
{successMessage && (
```

```
<div className="alert alert-success py-2 text-center">
```

```
  {successMessage}
```

```
</div>
```

```
)}
```

```
<form onSubmit={handleLogin}>
```

```
  {/* Username */}
```

```
<div className="mb-4">
```

```
<label className="form-label fw-semibold">Username</label>
```

```
<input
```

```
  type="text"
```

```
  className="form-control form-control-lg"
```

```
  placeholder="Enter your username"
```

```
  value={values.username}
```

```
  onChange={(e) =>
```

```
    setValues({ ...values, username: e.target.value })
```

```
  }
```

```
  required
```

```
</div>
```

```

{ /* Password */}

<div className="mb-4">

  <label className="form-label fw-semibold">Password</label>

```

```

<div className="position-relative">

  <input

    type={showPassword ? "text" : "password"}

    className="form-control form-control-lg pe-5"

    placeholder="Enter your password"

    value={values.password}

    onChange={(e) =>

      setValues({ ...values, password: e.target.value })

    }

    required

  />

```

```

{ /* Password Toggle */}

<button

  type="button"

  className="btn position-absolute top-50 end-0 translate-middle-

y me-2"

  onClick={() => setShowPassword(!showPassword)}

  style={{ border: "none" }}

>

```

```

        <i
            className={`bi ${showPassword ? "bi-eye-slash" : "bi-eye"}
fs-5`}
        ></i>
    </button>
</div>
</div>

{ /* Login Button */}
<div className="d-grid mt-4">
    <button
        type="submit"
        className="btn btn-success btn-lg fw-semibold"
        disabled={loggingIn}
    >
        {loggingIn && (
            <span className="spinner-border spinner-border-sm me-
2"></span>
        )}

        {loggingIn ? "Signing in..." : "Login to System"}
    </button>
</div>
</form>
</div>

```

</div>

</div>

</div>

</div>

{/* Footer */}

<div className="text-center py-3 small text-muted">

© {new Date().getFullYear()} National Primary Health Care Development
Agency

</div>

</div>

);

};

export default Login;

import React, { useMemo } from "react";

import { useOutletContext } from "react-router-dom";

const DfpDashboard = () => {

const { user } = useOutletContext();

// Format Last Login

const formattedLastLogin = user?.last_login

? new Date(user.last_login).toLocaleString()

: "First login";

```

return (
    <div className="container-fluid py-4" style={{ backgroundColor:
"#fff" }}>
        { /* Page Header */ }

        <div className="card border-0 shadow-sm mb-4">

            <div className="card-body d-flex justify-content-between align-items-
center flex-wrap">

                { /* Left Section */ }

                <div>

                    <h2 className="fw-bold mb-1">

                        <i className="bi bi-speedometer2 text-primary me-2"></i>

                        Dashboard

                    </h2>

                    <div className="text-muted">

                        <i className="bi bi-building me-2"></i>

                        {user?.department}

                    </div>

                    <small className="opacity-75 d-block mt-2">

                        <i className="bi bi-clock-history me-2"></i>

                        Last Login: {formattedLastLogin}

                    </small>

                </div>

                { /* Right Section (User Badge) */ }

```

```

<div className="d-flex align-items-center mt-3 mt-md-0">

  {/* Role Badge */}

  <div className="text-end">

    <span className="badge bg-secondary px-3 py-2">

      <i className="bi bi-person-badge-fill me-1"></i>

        Focal Person

      </span>

    </div>

  </div>

</div>

</div>

</div>

</div>

{/* Statistic Cards */}

<div className="row g-3 mb-4">

  <div className="col-md-3">

    <div className="card shadow-sm border-0">

      <div className="card-body d-flex align-items-center">

        <i className="bi bi-folder-fill fs-1 text-primary me-3"></i>

        <div>

          <h6 className="mb-1">Total Documents</h6>

          <h4 className="fw-bold">1,250</h4>

        </div>

      </div>

    </div>

  </div>

</div>

</div>

</div>

```



```

<div className="col-md-3">
  <div className="card shadow-sm border-0">
    <div className="card-body d-flex align-items-center">
      <i className="bi bi-clock-history fs-1 text-warning me-3"></i>
      <div>
        <h6 className="mb-1">Pending Requests</h6>
        <h4 className="fw-bold">32</h4>
      </div>
    </div>
  </div>
</div>
<div className="col-md-3">
  <div className="card shadow-sm border-0">
    <div className="card-body d-flex align-items-center">
      <i className="bi bi-person-lines-fill fs-1 text-success me-3"></i>
      <div>
        <h6 className="mb-1">Active Users</h6>
        <h4 className="fw-bold">120</h4>
      </div>
    </div>
  </div>
</div>
<div className="col-md-3">
  <div className="card shadow-sm border-0">
    <div className="card-body d-flex align-items-center">

```

<i className="bi bi-exclamation-circle-fill fs-1 text-danger me-3"></i>

<div>

<h6 className="mb-1">Expiring Documents</h6>

<h4 className="fw-bold">7</h4>

</div>

</div>

</div>

</div>

</div>

{/* Recent Activity */}

<div className="card shadow-sm border-0">

<div className="card-header bg-white d-flex justify-content-between align-items-center">

<h5 className="mb-0">Recent Activities</h5>

<small className="text-muted">Last 7 days</small>

</div>

<div className="card-body p-0">

<div className="table-responsive">

<table className="table table-hover mb-0">

<thead className="table-light">

<tr>

<th scope="col">Document</th>

<th scope="col">Uploaded By</th>

```

    <th scope="col">Department</th>

    <th scope="col">Status</th>

    <th scope="col">Date</th>

</tr>

</thead>

<tbody>

<tr>

    <td>Annual Report 2025</td>

    <td>Jane Doe</td>

    <td>PRSDFF</td>

    <td>

        <span className="badge bg-success">Approved</span>

    </td>

    <td>2026-02-16</td>

</tr>

<tr>

    <td>Policy Document</td>

    <td>John Smith</td>

    <td>Finance</td>

    <td>

        <span className="badge bg-warning text-dark">Pending</span>

    </td>

    <td>2026-02-14</td>

</tr>

<tr>

```

```
        <td>Partner MoU</td>
        <td>Mary Johnson</td>
        <td>Partners</td>
        <td>
            <span className="badge bg-danger">Expired</span>
        </td>
        <td>2026-01-30</td>
    </tr>
</tbody>
</table>
</div>
</div>
</div>
</div>
);
};
```

```
export default DfpDashboard;
```